



Agriculture and biodiversity damage: A prospective evaluation of the impact of Brazilian agriculture on its ecoregions through life cycle assessment methodology

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ABSTRACT

The natural ecosystems' replacement by farmland and the consequent biodiversity damage (BD) for agriculture are one of the principal concerns worldwide. The development of the life cycle assessment (LCA) methodology involves enormous efforts to include BD parameters and develop a prospective LCA approach for future evaluations of production and technologies use. Thus, this work aims to determine the current impacts and estimate the future impacts in terms of damage to biodiversity caused by land occupation by agricultural commodities produced in Brazil, such as coffee, corn, oranges, and sugar cane, for the six ecoregions present in the country—Amazon, Atlantic Forest, Caatinga, Cerrado, Pampas, and Pantanal—in the 20-year period from 2015 to 2035. For this and to search for hotspots, we applied the indicators proposed by Chaudhary and Books (2018), for inventories whose functional unit is production per m² of 1 kg of crop. Although the Cerrado is one of the ecoregions in which deforestation has advanced the most, it has the area/production ratio that has evolved the most. In contrast, Pampas and Caatinga, which are not seen as agricultural frontiers, increased their impacts. The most optimistic scenarios for the future have been those in regions considered agricultural frontiers; however, these are the regions where agriculture is more technologically developed, for example, coffee production in the Atlantic Forest and in the Cerrado. **The results indicate that the technological development of agriculture can contribute to mitigating the impacts of damage to biodiversity in the future**, and that the implementation of legislative and inspection measures is fundamental to supporting the correct use of the soil and preventing illegal soil change. Otherwise, in the future, we will see the increasing disappearance of species. Thus, we need researchers, farmers, and policy makers to move from development to conservation.

1. Introduction

The disappearance and fragmentation of natural ecosystems and, consequently, global biodiversity loss are two of the main current problems worldwide (Wolff et al., 2018). The destruction of tropical rainforest mostly contributed to the suggested sixth mass extinction (Ceballos et al., 2015). The disappearance of species does not only imply a genetic loss but also diminishes ecological actions and services that warrant, in addition to well-being, the maintenance of human life (Aerts et al., 2018; Caro et al., 2022; Jennings et al., 2016).

The Brazilian territory comprises six different biomes (ecoregions)

that are world biodiversity hotspots. All of them are arable and because of this, these ecoregions have suffered high degradation due to agricultural occupation and land use change. For example, the Atlantic Forest – the third largest biome in Brazil and the second largest tropical forest on the American continent – has historically been violated because of settlements and agricultural practices and is now one of the most devastated ecoregions on the planet (Lucas et al., 2021; Martins, 2011).

Agriculture, a pivotal activity for developing societies and human life maintenance (Fuller and Stevens, 2019), converts huge amounts of natural ecosystems into monocultures (Aznar-Sánchez et al., 2019).

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Although field crops do not sustain high biodiversity levels, agriculture depends for its existence on the ecosystem services provided by these species (Barrios et al., 2018). The damage resulting from biodiversity loss from land use change is even greater on tropical regions, which harbor the greatest numbers and variety of species in the entire planet (Harvey et al., 2020; Jablonski et al., 2006; Rolland et al., 2014).

Although encompassing different tropical ecoregions, Brazil is a country whose agribusiness was responsible for 27 % of the Brazilian gross domestic product (GDP) in 2021, mostly soybeans, maize, sugar, fruits, and meats, among many others (CNA, 2022; COMEX STAT, 2022). The aforementioned data predominantly refer to commodities and their supply chains, which represent 70 % of the agriculture production in the country (CNA, 2022). In Brazil's territory, the greatest biodiversity among all other countries is estimated—the first among the 18 megadiverse countries (Mittermeier et al., 2005; UNEP, 2019). About 1.8 million cataloged species (13.1 % of the total species on the planet) are described in its six ecoregions (Lewinsohn and Prado, 2005).

Hence, the country is one of the most important global food suppliers, a status which simultaneously poses a challenge to the planet's ecological and climate balance. Therefore, Brazilians yearn for the harmonious existence between agriculture and the preservation of natural areas, biodiversity, and ecosystem services, which are essential not just for them but also for the entire world. Investigations that allow the direct evaluation of the damage caused by farming to biodiversity are essential to finding information and solutions to mitigate such impacts and thereby ensuring environmental and economic sustainability.

Life cycle assessment (LCA) is considered one of the most promising methodologies for evaluating the environmental impacts of complete production systems (Hellweg and Milà i Canals, 2014; van der Giesen et al., 2020). Furthermore, this methodology is recognized by the International Organization for Standardization (ISO), which allows the use of evaluations for international environmental certifications (ISO 14040, 2006a, 2006b; ISO 14044, 2006). As contemporary agriculture is composed of highly complex production systems assuming industrial proportions (Alexandratos and Bruinsma, 2012), and destined to supply international markets, LCA comes up as a promising methodology for the agricultural sector (Alhashim et al., 2021).

In recent years, the LCA methodology has expanded its range of impact assessments, including biodiversity loss, ecological functions, and ecosystem services. Biodiversity loss caused by land use change because of agriculture is one of the categories considered most important for improving the methodology (Lucas et al., 2021). Thus, the UNEP/SETAC Life Cycle Initiative has developed a guide that includes past efforts and new challenges to model these impacts (UNEP, 2016), using the set of indicators presented in Chaudhary et al. (2015). However, the characterization factors (CF) by Chaudhary et al. (2015) were updated by Chaudhary and Books (2018), modifying part of their values between publication.

Another improvement and expansion of applicability sought for the methodology was to allow the application of a new perspective: the prospective LCA. This approach has the purpose of evaluating the environmental performance of technologies (or a set of them, such as agricultural production systems) that are being used and may have expanded use in the future, in early or advanced development such as a forecast of impacts (Arvidsson et al., 2018; Thonemann et al., 2020). However, this new approach still depends on studies and evaluations that lead the way to its improvement, which will allow prospective evaluations to be more accurate and actually contribute to mitigating future impacts. The indicators of biodiversity loss have not previously been applied in a LCA study with a prospective approach. So, this study also intends to contribute to the LCA methodology.

The combination of agricultural performance and biodiversity loss evaluated by the latest advances is considered one of the most promising methodologies for evaluating agriculture's environmental impacts. So, an important question addressed in this study is: how can we use the LCA methodology to delineate a more sustainable agriculture in terms of

biodiversity?

Therefore, the main objective of this article is to investigate the impact of some agricultural commodities produced in the different ecoregions of Brazil on the biodiversity of such regions and make future projections of the increase or reduction of these impacts. It aims to find the worst-case scenarios and discuss solutions in terms of how agriculture may mitigate biodiversity damage while continuing to fulfill its role. Finally, we investigated whether the potential biodiversity damage (BD) indicators selected here are adequate to generate a prospective LCA study and could help predict biodiversity loss due to agriculture. Thus, this paper represents an empirical and methodological contribution.

2. Methods

2.1. Crops

We consider the following crops and their impacts for this study: coffee (only *Coffea arabica*), maize, orange, and sugarcane. The choice of crops was because of Brazil's leadership in production: the country was the world's largest producer of coffee, orange, and sugarcane, as well as the third largest producer of maize in 2019 (FAO, 2021). In 2020, the coffee production chain was responsible for 5.49 % of the country's total exports; cereals and flours (including maize), 6.78 %; juices and fruits (including orange), 1.59 % and 1 %; and sugarcane, 9.88 % (MAPA, 2021). Furthermore, except coffee, all these crops are grown in each of the six Brazilian ecoregions: Amazon, Atlantic Forest, Caatinga, Cerrado, Pampas, and Pantanal. Coffee is not grown in the Pampas and Pantanal ecoregions.

2.2. Prospective LCA

There is still no consensus on the guidelines for a prospective LCA study. However, a prospective study must first define the *status quo* of the technology or activity to be evaluated. It should then build predictive scenarios. Finally, the study should include extreme scenarios, which can be obtained by applying stoichiometric relationships, in order to analyze the best scenario (Arvidsson et al., 2018). The major challenges to conducting a prospective LCA are comparability, data source, and uncertainties (Thonemann et al., 2020). Thus, for this study, these recommendations and limitations are considered and presented in more detail in Section 2.2.1 and in Fig. 1.

2.2.1. Elaborating prospective scenarios

In this study, the *status quo* value is the area needed to produce 1 kg of crop in 2015. To determine the prospective scenarios, it was first necessary to determine the difference in area/production proportion for crops (APC) (% of area/production altered) between 2015 and 2020. The APC of this study were determined using Eq. (1).

$$AP_C = AP_{2020} - AP_{2015} \text{ or } AP_C = 1 - (AP_{2020}/AP_{2015}) \quad (1)$$

where AP_C = difference of APC in %; AP_{2015} = area of crop production in m^2 /production of 1 kg of yield for 2015; and AP_{2020} = area of crop production in m^2 /production of 1 kg of yield for 2020.

To determine the prospective scenarios (SC_p), Eq. (2) was applied.

$$SC_{py} = AP_y + (AP_y \cdot AP_{DC}) \quad (2)$$

where SC_p = prospective scenarios, in m^2 /production of 1 kg of yield; AP = area of crop production, in m^2 /production of 1 kg of yield; and AP_{DC} = area of crop production, in m^2 /production of 1 kg of crop (in %); y = year (determined or estimated).

As the occupation and production of crops were initially determined for a period of five years, from 2015 to 2020, the same proportions were adopted for the development of prospective scenarios until 2035 (Fig. 1).

There are two key assumptions that underlie the results of this paper:

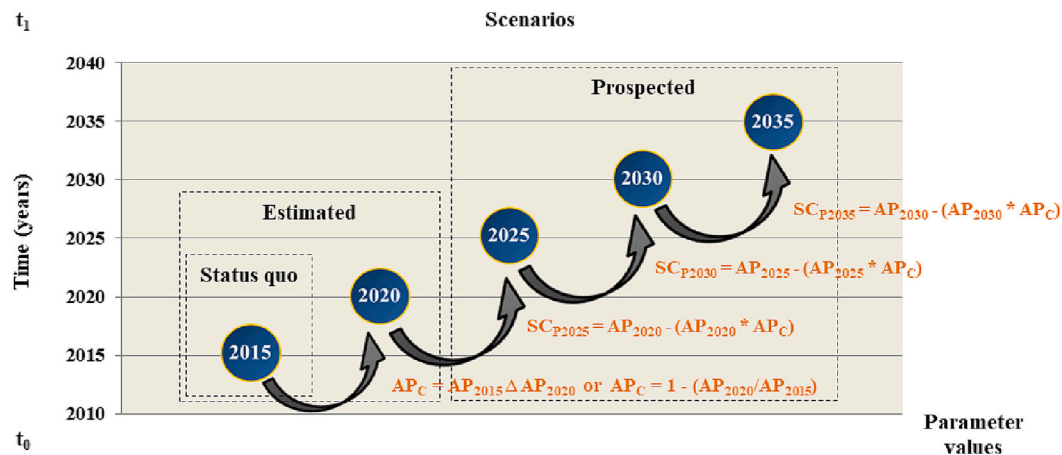


Fig. 1. Evolution and determination of estimated (future) scenarios to determine damage to biodiversity by agricultural crops, where APC = difference of APC (in %); AP2015 = area of crop production in m²/production of 1 kg of crop for the 2015 year; AP2020 = area of crop production in m²/production of 1 kg of crop for the 2020 year; SCP = predictive scenarios, in m²/production of 1 kg of crop; APy = area of crop production, in m²/production of 1 kg of crop; y = year (determined or estimated).

i) continuous demand for agricultural products; ii) continuous technology, so we can linearly extrapolate hectare-increases of yields. Although restrictive, these kinds of assumptions are common for modeling.

2.3. Scenarios

2.3.1. Baselines

The baselines, values, and functional units were determined by data from the Brazilian Institute of Geography and Statistics (IBGE): Municipal Agricultural Production (PAM) (IBGE, 2021), from which the values of harvested area and the amount produced of each crop in all Brazilian municipalities ($n = 5533$ for each year) were taken. For more information on how the values and functional units were determined for the estimated scenarios, see supporting information, Tables S2 (2015 Data) and S3 (2020 Data). Table 1 summarizes the estimated functional units for 2015 and 2020 for each crop in the six ecoregions.

2.3.2. Prospective scenarios

The prospective scenarios were performed for 2025, 2030, and 2035. Their functional units were determined by applying Eq. (2). The prospective scenarios are evolutionary, that is, the 2025 scenario follows the variation of the 2020 area/production ratio, and so on. For more information on how the functional units of the prospected scenarios

were determined, see the supporting information in Table S4 (prospective values). Table 2 presents the values of the prospective functional units for 2025, 2030, and 2035 for each crop in the six ecoregions.

2.4. Indicators of potential biodiversity damage (BD)

To evaluate the potential damage to biodiversity in Brazil, we applied the indicators proposed by Chaudhary and Books (2018), the updated version of Chaudhary et al. (2015). These are the indicators recommended by the United Nations Environment Program (UNEP) and the Society of Environmental Toxicology and Chemistry (SETAC) in the Global Guidance for Life Cycle Impact Assessment Indicator (UNEP, 2016).

The CF of Chaudhary and Books (2018) represent the damage to biodiversity resulting from global land use in potentially disappeared fraction (PDF) in year/m², for five taxa: mammals, birds, amphibians, reptiles, and plants, in addition to aggregated values of taxa. In addition to damages for the five different types of land use (managed forests, plantations, pastures, crops, and urban), at three levels of intensity (minimal, light, and intense), for each of the 804 existing and classified terrestrial ecoregions.

The CFs were based on the Countryside SAR model (Pereira et al., 2014), which classifies species into functional groups with particular

Table 1

Values of Functional Units (FU), in m²/production of 1 kg of crop, determined for the 2015 and 2020 years for four crops (coffee, maize, orange, and sugarcane) cultivated in six ecoregions in Brazil (Amazon, Atlantic Forest, Caatinga, Cerrado, Pampas, and Pantanal).

Crops	Ecoregions/code ^a					
	Amazon	Atlantic Forest	Caatinga	Cerrado	Pampas	Pantanal
	NT0168	NT0150	NT1304	NT0704	NT0710	NT0907
m ² /production of 1 kg of crop						
2015						
Coffee	13.37	6.34	10.35	7.11	0.00	0.00
Maize	1.80	1.43	5.17	1.55	1.65	1.57
Orange	0.65	0.40	0.74	0.37	1.08	0.81
Sugarcane	0.145	0.134	0.213	0.128	0.26	0.131
2020						
Coffee	9.36	5.06	6.63	4.93	0.00	0.00
Maize	1.70	1.64	4.05	1.52	1.75	1.63
Orange	0.47	0.34	0.85	0.31	0.96	1.17
Sugarcane	0.125	0.133	0.154	0.127	0.276	0.128

Source: WWF (2021).

^a The code represents the classification of ecoregions by the World Wildlife Fund (WWF) ecoregion code list.

Table 2

Values of functional units (FU), in m²/production of 1 kg of crop, prospected for 2025, 2030, and 2035 for four crops (coffee, maize, orange, and sugarcane) cultivated in six ecoregions present in Brazil (Amazon, Atlantic Forest, Caatinga, Cerrado, Pampas, and Pantanal).

	Ecoregions/code ^a					
	Amazon	Atlantic Forest	Caatinga	Cerrado	Pampas	Pantanal
	NT0168	NT0150	NT1304	NT0704	NT0710	NT0907
Crops	m ² /production of 1 kg of crop					
2025						
Coffee	6.56	4.03	4.25	3.42	0.00	0.00
Maize	1.61	1.88	3.17	1.48	1.86	1.68
Orange	0.34	0.29	0.98	0.25	0.86	1.69
Sugarcane	0.108	0.132	0.112	0.126	0.293	0.125
2030						
Coffee	4.59	3.21	2.72	2.37	0.00	0.00
Maize	1.52	2.16	2.49	1.45	1.98	1.73
Orange	0.25	0.25	1.12	0.21	0.77	2.45
Sugarcane	0.094	0.13	0.081	0.125	0.311	0.121
2035						
Coffee	3.21	2.56	1.75	1.64	0.00	0.00
Maize	1.43	2.47	1.95	1.42	2.10	1.79
Orange	0.18	0.21	1.29	0.17	0.69	3.54
Sugarcane	0.081	0.129	0.059	0.124	0.331	0.118

Source: WWF (2021).

^a The code represents the classification of ecoregions by the World Wildlife Fund (WWF) ecoregion code list.

affinities to different habitats. There are two types of CFs: regional and global. To differentiate them, consider the presence of endemic species in an ecoregion: their loss is translated into global loss of species (extinction), so CF_{Global} considers one more parameter in its equation, the vulnerability score (VS) (Chaudhary et al., 2015).

The VS represents changes in the relative abundance of species in the ecoregion, which is based on the endemic richness (interpreted as the region-specific contribution to global biodiversity) of each taxon in each ecoregion. It is estimated by multiplying the range fraction of each species with the threat level assigned by the International Union for Conservation of Nature (IUCN, 2017), thus calculating the threatened endemic richness. Finally, VS is defined as the ratio of threatened endemic richness to total species richness, thus representing the level of threat to the species on a global scale (Chaudhary et al., 2015).

For this study, we only assessed potential damage to global biodiversity (BD_{Global}) impacts (Chaudhary and Books (2018), Eq. (3), and not regional ones because the products are commodities of interest with global impacts.

$$BD_{Global,g} = CF_{Global,g,i,j} \cdot A_{i,j} \quad (3)$$

where BD = Potential biodiversity damage, in PDF × year m²; g = taxonomic group (taxa); CF_{Global} = global characterization factors, in potentially disappeared fraction (PDF) × year m²; i = land use; j = ecoregion; A = area total of crop, in m².

Table 3

Characterization of factors global (CF_{Global}) values of ecoregions for each taxa, in potentially disappeared fraction (PDF) × year/m², for cropland intense use.

Biomes	Code ^{a,1}	Taxa ²				
		Mammals	Birds	Amphibians	Reptiles	Plants
		CF _{Global}				
Amazon	NT0168	7.23E-12	1.40E-11	6.08E-12	3.68E-13	5.23E-11
Atlantic Forest	NT0150	1.64E-11	2.75E-11	3.24E-07	1.13E-12	4.15E-11
Caatinga	NT1304	4.58E-12	7.21E-12	6.01E-12	1.01E-12	3.72E-12
Cerrado	NT0704	5.25E-12	7.31E-12	7.97E-12	1.09E-12	7.51E-12
Pampas	NT0710	8.74E-12	9.25E-12	8.83E-12	9.76E-13	1.66E-11
Pantanal	NT0907	4.42E-12	6.91E-12	5.05E-12	7.59E-13	1.39E-117

Source: ¹WWF (2021); ²Chaudhary and Books (2018).

^a The code represents the classification of ecoregions by the World Wildlife Fund (WWF) ecoregion code list.

2.4.1. Values of characterization factors (CF) for ecoregions

Regarding the intensity level of the crops, intense agricultural practices were considered (confidence interval values of 95 %). “Intense” level was adopted because of the high yield of crops in Brazil, adjusted to the criteria defined by Chaudhary and Books (2018): large areas, application of inorganic fertilizers, application of pesticides, irrigation, mechanization, and absence of crop rotation. The CF_{Global} values are summarized in Table 3.

2.5. Life cycle inventory

The functional unit is responsible for representing the quantified performance of a product system for use as a reference unit (ISO 14044, 2006). Therefore, as the functional unit of the indicators is potential biodiversity damage, in PDF × year m², we adopted the functional unit of m²/production of 1 kg of crop. In Fig. 2, we show how each stage of the study connects and where the functional unit fits.

3. Results

The results of the assessment of impacts on biodiversity in the Amazon, Atlantic Forest, Caatinga, Cerrado, Pampas, and Pantanal ecoregions for each crop (coffee, maize, orange, and sugarcane) for estimated and prospective scenarios are shown in Figs. 3–8. The

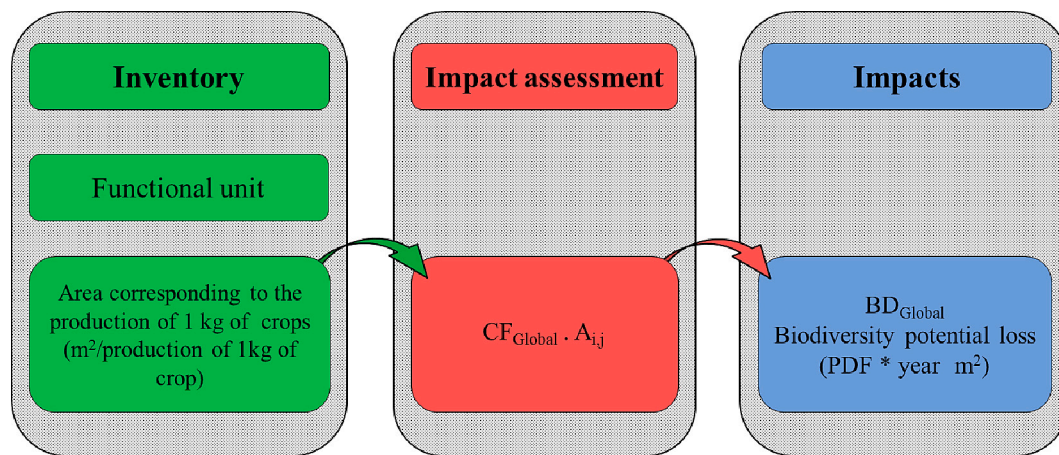


Fig. 2. Flowchart of life cycle inventory and of the potential biodiversity damage (BD) assessment in the production of crops in Brazil (coffee, maize, orange, and sugarcane), where CF_{Global} = global characterization factors, in potentially disappeared fraction (PDF) $\times$ year m^2 ; i = land use; j = ecoregion; A = area total of crop, in m^2 .

potential damage was characterized for the five taxa—plants, birds, mammals, reptiles, and amphibians.

3.1. Amazon

A reduction in the impacts of BD between five-year periods was observed for coffee cultivation, which is concentrated in Rondônia State. For the estimated scenarios, a significant reduction in impacts of 42.8 % was observed. The calculated prospective scenarios indicate that reduction of impacts would be 315.82 % between 2015 and 2035. A reduction in both yields and cultivated area was observed. Between 2015 and 2020, production reduction was $4.80E+04$ kg (approximately 69 %) and $8.30E+05$ m^2 (approximately 65 %). Thus, the area/production ratio of coffee changed from 13.37 in 2015 to 9.36 in 2020 and could reach 3.21 m^2 /production in 2035.

However, the area/production ratio has been mainly affected by the reduction of the cultivated area rather than increasing production. Reducing impacts by diminishing cultivated area, especially when production declines simultaneously, indicates neither technical development, nor crop economic viability. Thus, mitigating the impacts of damage to biodiversity by reducing the extension of cropland may be an unsustainable alternative because of replacement by other crops, such as soybean or livestock, which may impact in the same way or even more intensely.

A reduction in the impacts of BD on maize cultivation between the five-year periods was observed. In the estimated scenarios, the reduction in impacts was 5.87 %. In the prospective scenarios, the reduction may be 25.62 % between 2015 and 2035. Between 2015 and 2020, production increased by $1.22E+10$ kg (approximately 69 %) and areas by $1.22E+10$ m^2 (approximately 59 %). Thus, the area/production ratio of maize changed from 1.8 in 2015 to 1.7 in 2020 and could reach 1.43 m^2 /production in 2035.

The increase in maize grain production was impressive, possibly because of technical advances in cultivation, although the increase in the amount of cultivated area reduces the merit of reducing the impact of cultivation. The equations to create the prospective scenarios continue to consider both the increase in production and the amount of cultivated area, and even then, the results are positive. If maize cultivation in this ecoregion maintains increasing production and reduces the expansion of cultivated areas, the damage to biodiversity could be even smaller.

Orange cultivation has shown reduction in the impacts of BD between the five-year periods. In the estimated scenarios, the reduction in impacts was 5.87 %. In the prospective scenarios, the reduction may be 25.62 % between 2015 and 2035. Between 2015 and 2020, the increase

in production was $1.19E+08$ kg (approximately 37 %) and the reduction was $1.71E+05$ m^2 (approximately 0.1 %). Thus, the area/production ratio of orange changed from 0.65 in 2015 to 0.47 in 2020 and could reach 0.18 m^2 /production in 2035.

For sugarcane cultivation too, reduction in the impacts of damage to biodiversity between the five-year periods was also observed. In the estimated scenarios, the reduction in impacts was 15.62 %, and for the prospective scenarios, the reduction could be 78.67 % between 2015 and 2035. As observed for the orange crop, sugarcane production increased production cultivated area was reduced. Between 2015 and 2020, production increase was $2.02E+09$ kg (approximately 13 %) and area reduction was $5.01E+07$ m^2 (approximately 2 %). Thus, the area/production ratio of sugarcane changed from 0.145 in 2015 to 0.125 in 2020 and could reach 0.081 m^2 /production of 1 kg of sugarcane in 2035. As observed for orange, impacts for sugarcane cultivation in the ecoregion were reduced through the best possible scenario: increased production and reduced cultivated area.

3.2. Atlantic forest

The reduction of impacts for coffee cultivation (25.44 %), which is concentrated in this ecoregion, was estimated for a five-year period (Fig. 5). For the prospective scenarios, the impact reduction could be 147.62 % between 2015 and 2035. Increasing yields and a reduction of extension cultivation led to this result in 2020, yields increased by $3.94E+08$ kg (approximately 20 %) and $5.23E+08$ m^2 (approximately 4 %). The area/production ratio diminished from 6.34 in 2015 to 5.06 in 2020 and could reach 2.56 m^2 /production in 2035. These simulations indicate the best possible scenario: increasing production while simultaneously reducing cultivated areas, so the future environmental benefit pointed out in this study is viable.

For maize, increasing damage (12.73 %) between the five-year periods was estimated. In prospective scenarios, increases could reach 41.98 % between 2015 and 2035. Production diminished and area extension increased: $3.75E+09$ kg (approximately 9 %) and $2.75E+09$ m^2 (approximately 4 %). The area/production ratio increased from 1.43 in 2015 to 1.64 in 2020 and could eventually reach 2.47 in 2035. These data indicate the worst-case scenario: production reduction and increasing cultivated area.

Impact reduction (17.03 %) for orange cultivation was characterized. For the prospective scenarios, the reduction could reach 87.59 %. Increasing fruit productions and simultaneous diminishing cultivated area: $3.91E+08$ kg of fruits (approximately 3 %) and reduction of $7.74E+08$ m^2 (approximately 12 %) led to area/production ratio reduction from 0.4 to 0.34 and could reach 0.21 in 2035. The reduction

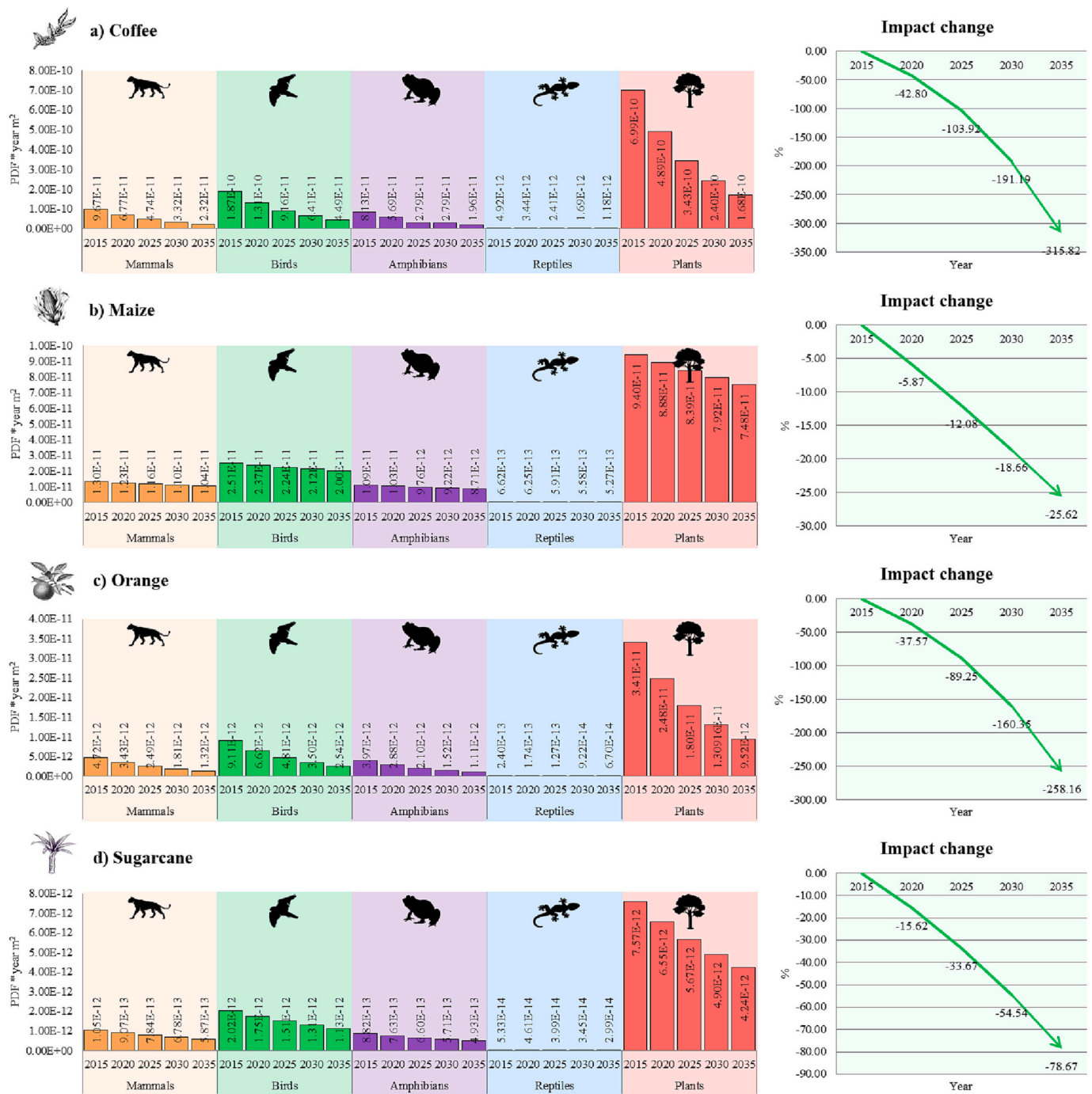


Fig. 3. Impact of potential global biodiversity damage (BD_{Global}), in potentially disappeared fraction (PDF) $\times$ year m^2 , for coffee, maize, orange, and sugarcane crops in the Amazon ecoregion for five taxa (mammals, birds, amphibians, reptiles, and plants). The information on the right side of the graphs indicate the increase or decrease (change) in the impacts of different scenarios.

of impacts was mainly because of the reduction of cultivated area, a factor that may compromise the permanence of the cultivar in the region in the future.

For sugarcane, reduction damage (1.01 %) between the five-year periods was estimated. In the prospective scenarios, reduction could reach 4.12 % between 2015 and 2035. Production and area were diminished: 3.20E+09 kg (approximately 1 %) and 1.29E+09 m^2 of area (approximately 2 %). The area/production ratio increased from 0.134 in 2015 to 0.133 in 2020 and could eventually reach 0.129 in 2035. These data indicate the pessimistic scenario: reduction in production and cultivated area.

3.3. Caatinga

The reduction of impacts for coffee cultivation (56.05 %), which is concentrated in this ecoregion, was estimated for a five-year period (Fig. 6). For the prospective scenarios, the impact reduction could be 493.03 % between 2015 and 2035. Reducing both production and extension cultivation led to this result in 2020: 1.20E+07 kg (approximately 13 %) and 4.25E+08 m^2 (approximately 44 %). The area/production ratio diminished from 10.35 in 2015 to 6.63 in 2020 and could reach 1.75 m^2 /production in 2035. These simulations indicate that there is no real reduction in impacts, but there is drastic reduction in the

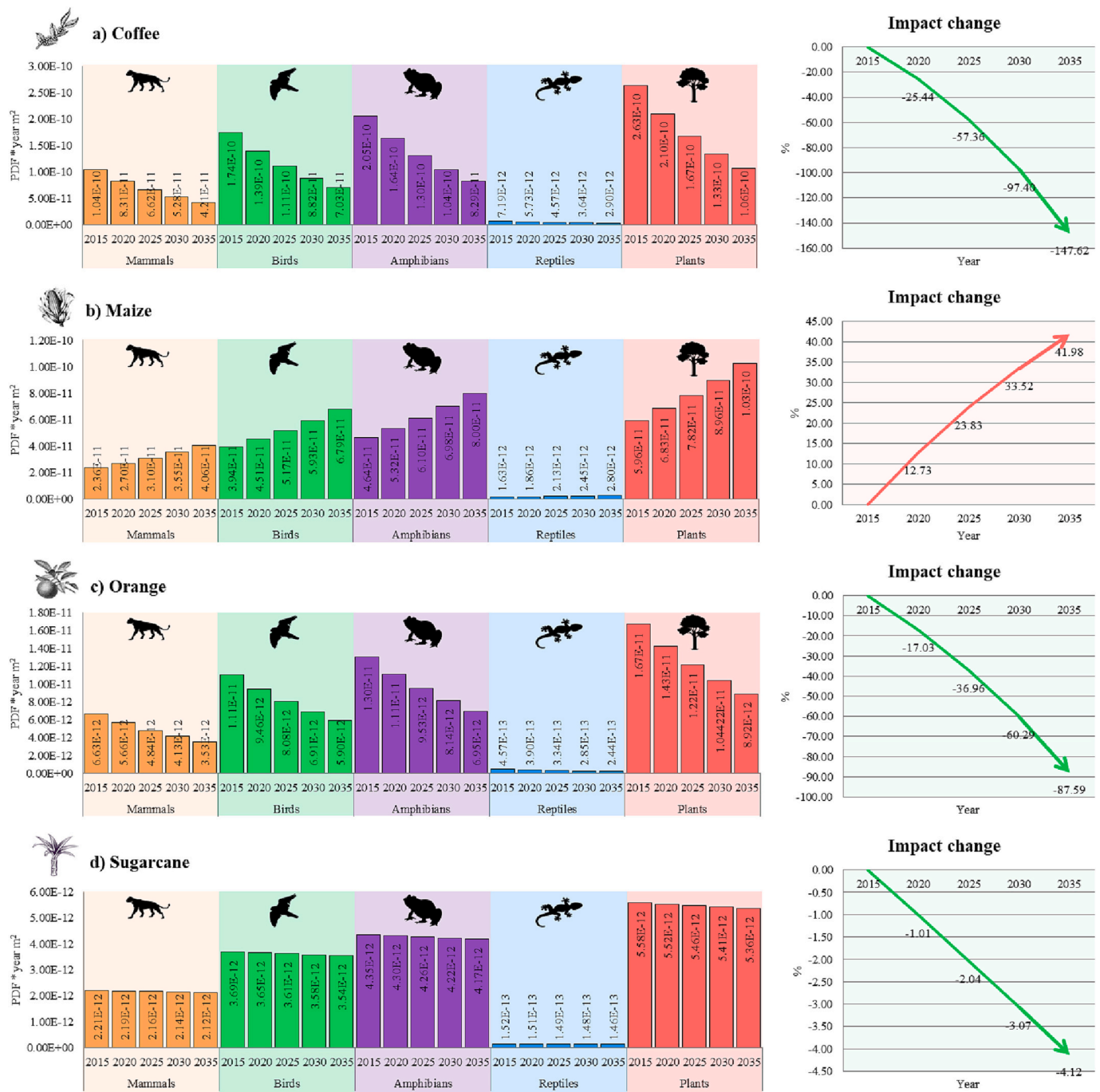


Fig. 4. Impact of potential biodiversity damage global (BD_{global}), in potentially disappeared fraction (PDF) $\times$ year m^2 , for coffee, maize, orange, and sugarcane crops in the Atlantic Forest ecoregion for five taxa (mammals, birds, amphibians, reptiles, and plants). The information on the right side of the graphs indicate the increase or decrease (change) in the impacts of different scenarios.

presence of the crop in the ecoregion.

Reduction in the impacts of BD on maize cultivation between the five-year periods was observed. In the estimated scenarios, the reduction in impacts was 27.57 %. In the prospective scenarios, the reduction may be 164.88 % between 2015 and 2035. Between 2015 and 2020, production increased to $1.27E+09$ kg (approximately 42 %) and area increased to $1.84E+09$ m^2 (approximately 12 %). Thus, the area/production ratio of maize changed from 5.17 in 2015 to 4.05 in 2020 and could reach 1.95 m^2 /production in 2035. If maize cultivation in this ecoregion maintains increasing production and reduces the expansion of cultivated areas, the damage to biodiversity could be even smaller.

Orange cultivation has shown increase in the impacts of BD between

the five-year periods. In the estimated scenarios, the increase in impacts was 12.95 %, while for the prospective scenarios the reduction may be 42.57 % between 2015 and 2035. Between 2015 and 2020, the increase in production was $4.09E+08$ kg (approximately 40 %) and the reduction in production was $2.34E+08$ m^2 (approximately 31 %). Thus, the area/production ratio of orange changed from 0.74 in 2015 to 0.85 in 2020 and could reach 1.29 m^2 /production in 2035. The reduction of impacts was observed through the best possible scenario: significant production increase and almost stable reduction in cultivated area.

For sugarcane cultivation, reduction in the impacts of damage to biodiversity between the five-year periods was also observed. In the estimated scenarios, the reduction in impacts was 37.94 % and for the

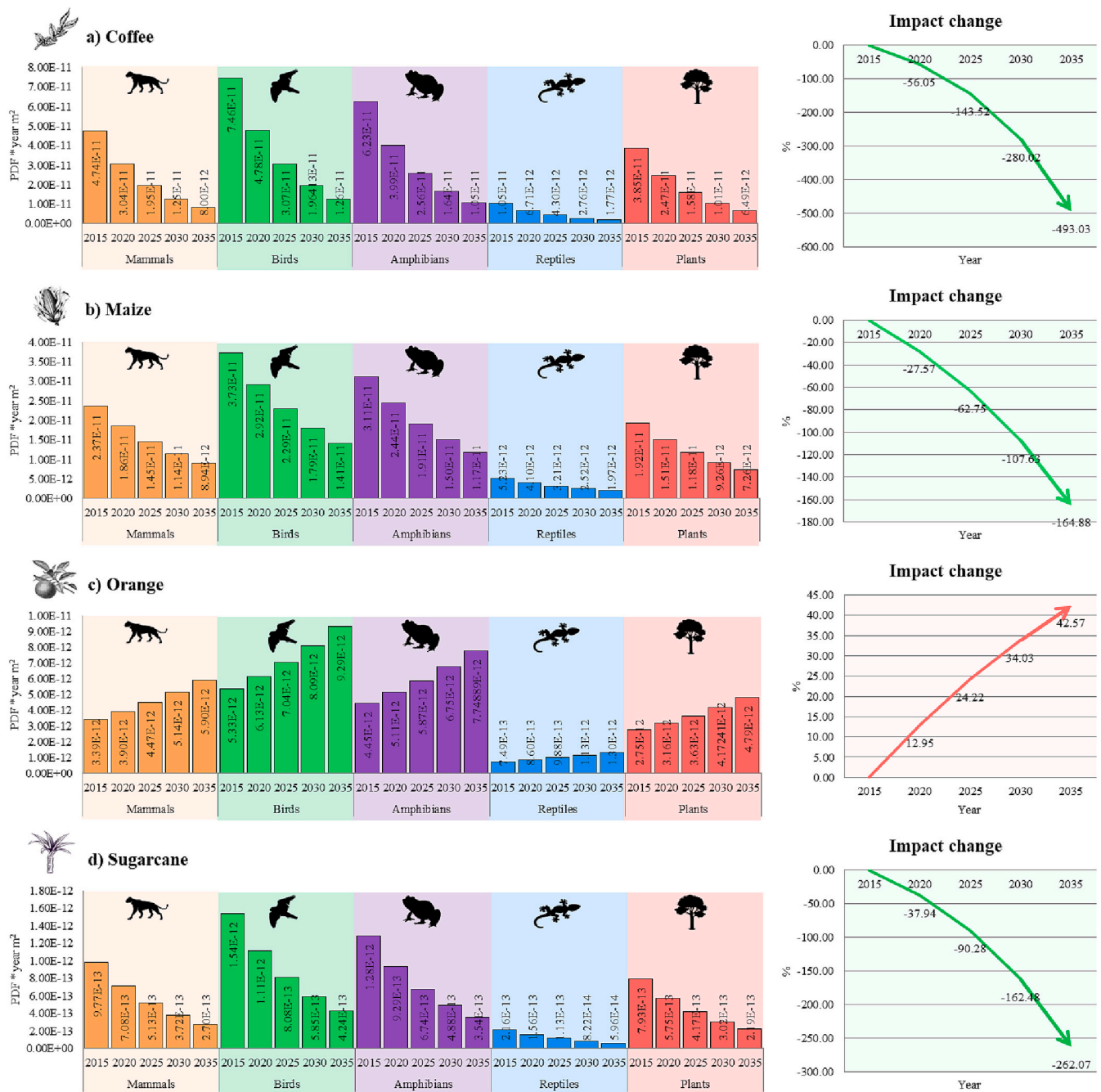


Fig. 5. Impact of potential biodiversity damage global (BD_{global}), in potentially disappeared fraction (PDF) $\times$ year m^2 , for coffee, maize, orange, and sugarcane crops in the Caatinga ecoregion for five taxa (mammals, birds, amphibians, reptiles, and plants). The information on the right side of the graphs indicate the increase or decrease (change) in the impacts of different scenarios.

prospective scenarios the reduction could be 262.07 % between 2015 and 2035. The crop reduced both production and cultivated area. Between 2015 and 2020, production reduction was $1.63E+09$ kg (approximately 15 %) and reduction of area was $8.94E+08$ m^2 (approximately 38 %). Thus, the area/production ratio of sugarcane changed from 0.213 in 2015 to 0.154 in 2020 and could reach 0.059 m^2 /production in 2035. The sugarcane scenarios indicate that there was no real reduction in impacts, but there was relevant reduction in the cultivated area.

3.4. Cerrado

For coffee, in the estimated scenarios, a significant reduction in

impacts of 44.1 % was observed. The calculated prospective indicates that the reduction of impacts would be 332.03 % between 2015 and 2035. An increase in both production and cultivated area was observed. Between 2015 and 2020, yields increased by $4.89E+08$ kg of grain (approximately 63 %) and $7.26E+08$ m^2 (approximately 13 %). Thus, the area/production ratio of coffee changed from 7.11 in 2015 to 4.93 in 2020 and could reach 1.64 m^2 /production in 2035. The reduction of coffee impacts has occurred through a good scenario: expressive increase in production and increase in cultivated area lower than production.

A reduction in the impacts of BD on maize cultivation between the five-year periods was observed. In the estimated scenarios, the reduction in impacts was 2.09 %. In the prospective scenarios, the reduction may

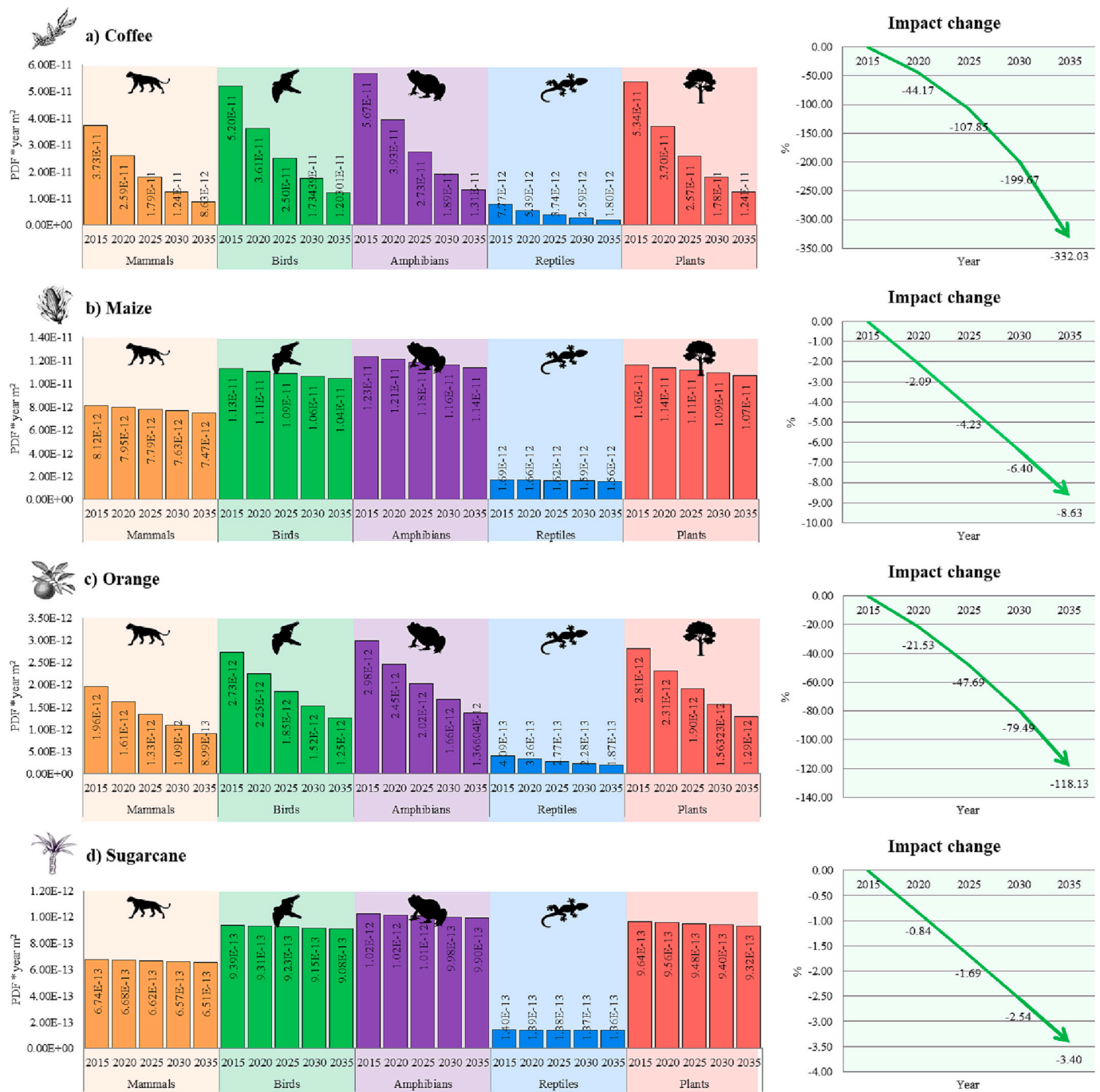


Fig. 6. Impact of potential biodiversity damage global (BD_{global}), in potentially disappeared fraction (PDF) $\times$ year m^2 , for coffee, maize, orange, and sugarcane crops in the Cerrado ecoregion for five taxa (mammals, birds, amphibians, reptiles, and plants). The information on the right side of the graphs indicate the increase or decrease (change) in the impacts of different scenarios.

be 8.63 % between 2015 and 2035. Between 2015 and 2020, production increased $1.64E+10$ kg of grain (approximately 31 %) and $2.32E+10 m^2$ (approximately 28 %). Thus, the area/production ratio of maize changed from 1.55 in 2015 to 1.52 in 2020 and could reach $1.42 m^2$ /production in 2035. The increase in grain production was expressive, although the increase in cultivated area is almost proportional to production.

Orange cultivation has shown reduction in the impacts of BD between the five-year periods. In the estimated scenarios, the increase in impacts was 21.53 %, whereas for the prospective scenarios, the reduction may be 118.13 % between 2015 and 2035. Between 2015 and 2020, the increase in production was $1.25E+09$ kg (approximately 13

%) and the reduction was $2.73E+08 m^2$ (approximately 7 %). Thus, the area/production ratio of orange changed from 0.37 in 2015 to 0.31 in 2020 and could reach $0.17 m^2$ /production in 2035.

For sugarcane cultivation, the reduction in the impacts of damage to biodiversity between the five-year periods was also observed. In the estimated scenarios, the reduction in impacts was 0.84 %, and for the prospective scenarios, the reduction could be 3.4 % between 2015 and 2035. The crop increased both production and cultivated areas. Between 2015 and 2020, the production reduction was $1.87E+10$ kg (approximately 4 %) and $1.91E+09 m^2$ of area (approximately 3 %). Thus, the area/production ratio of sugarcane changed from 0.28 in 2015 to 0.27 in 2020 and could reach $0.24 m^2$ /production in 2035.

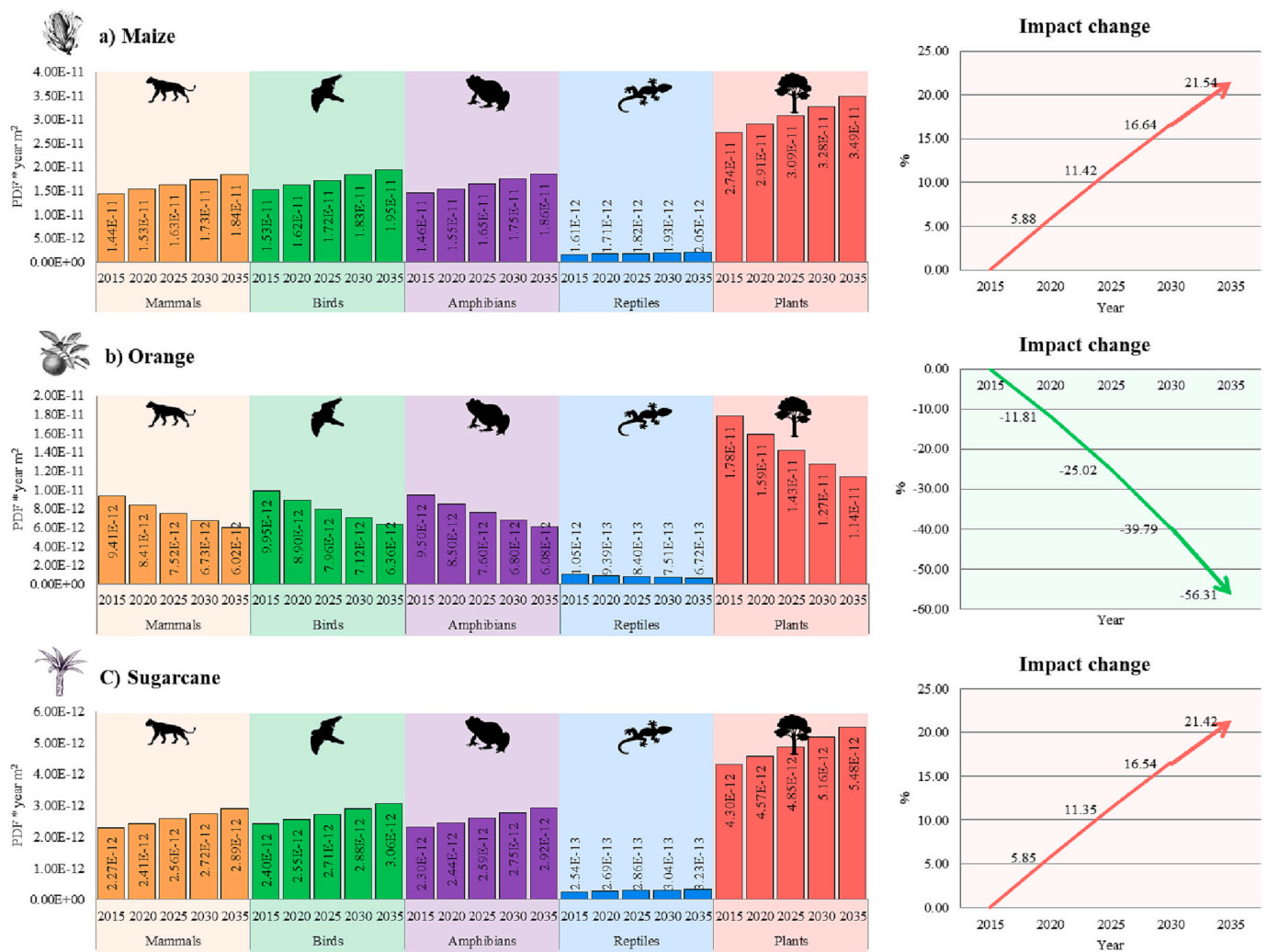


Fig. 7. Impact of potential biodiversity damage global (BD_{Global}), in potentially disappeared fraction (PDF) × year m², for maize, orange, and sugarcane crops in the Pampas ecoregion for five taxa (mammals, birds, amphibians, reptiles, and plants). The information on the right side of the graphs indicates the increase or decrease (change) in the impacts of different scenarios.

3.5. Pampas

The maize's increasing damage (5.88 %) between the five-year periods was estimated. In prospective scenarios, increases could reach 21.54 % between 2015 and 2035. Both production and area were diminished: 5.53E+08 kg of grain (approximately 20 %) and 6.87E+08 m² of area (approximately 15 %). The area/production ratio increased from 1.65 in 2015 to 1.75 in 2020 and could eventually reach 2.1 in 2035. The increase in the impacts of maize cultivation in the Pampas has been because of the reduction in production rather than the reduction in the amount of cultivated area.

Orange cultivation showed reduction in the impacts of BD between the five-year periods. In the estimated scenarios, the reduction in impacts was 11.81 %, whereas for the prospective scenarios, the reduction may be 56.31 % between 2015 and 2035. Between 2015 and 2020, the increase in production was 1.84E+07 kg of fruit (approximately 17 %) and 2.99E+07 m² of area (approximately 26 %). Thus, the area/production ratio of orange changed from 1.08 in 2015 to 0.96 in 2020 and could reach 0.69 m²/production in 2035. The evaluated scenarios indicate that there has been no real reduction in impacts, but an influence on the reduction of the area.

For sugarcane cultivation, increase in the impacts of damage to biodiversity between the five-year periods was also observed. In the estimated scenarios, the increase in impacts was 15.62 %, and for the

prospective scenarios, the increase could be 21.42 % between 2015 and 2035. Between 2015 and 2020, the reduction in production was 1.14E+08 kg (approximately 34 %) and 2.60E+07 m² of area (approximately 30 %). Thus, the area/production ratio of sugarcane changed from 0.26 in 2015 to 0.276 in 2020 and could reach 0.331 m²/production of 1 kg of sugarcane in 2035. The increase in impacts on sugarcane cultivation in the Pampas was because of the reduction in production rather than the reduction in the amount of cultivated area.

3.6. Pantanal

Increase in the impacts of BD on maize cultivation between the five-year periods was observed. In the estimated scenarios, the reduction in impacts was 3.18 %. In the prospective scenarios, the reduction may be 12.13 % between 2015 and 2035. Between 2015 and 2020, the reduction in production was 2.52E+07 kg of grain (approximately 2 %) and the increase was 2.37E+07 m² of area (approximately 1 %). Thus, the area/production ratio of maize changed from 1.57 in 2015 to 1.63 in 2020 and could reach 1.79 m²/production in 2035. The increase in impacts on maize cultivation in the Pantanal was because of the reduction in production and the increase in the amount of cultivated area.

Orange cultivation has shown an increase in the impacts of BD between the five-year periods. In the estimated scenarios, the increase in

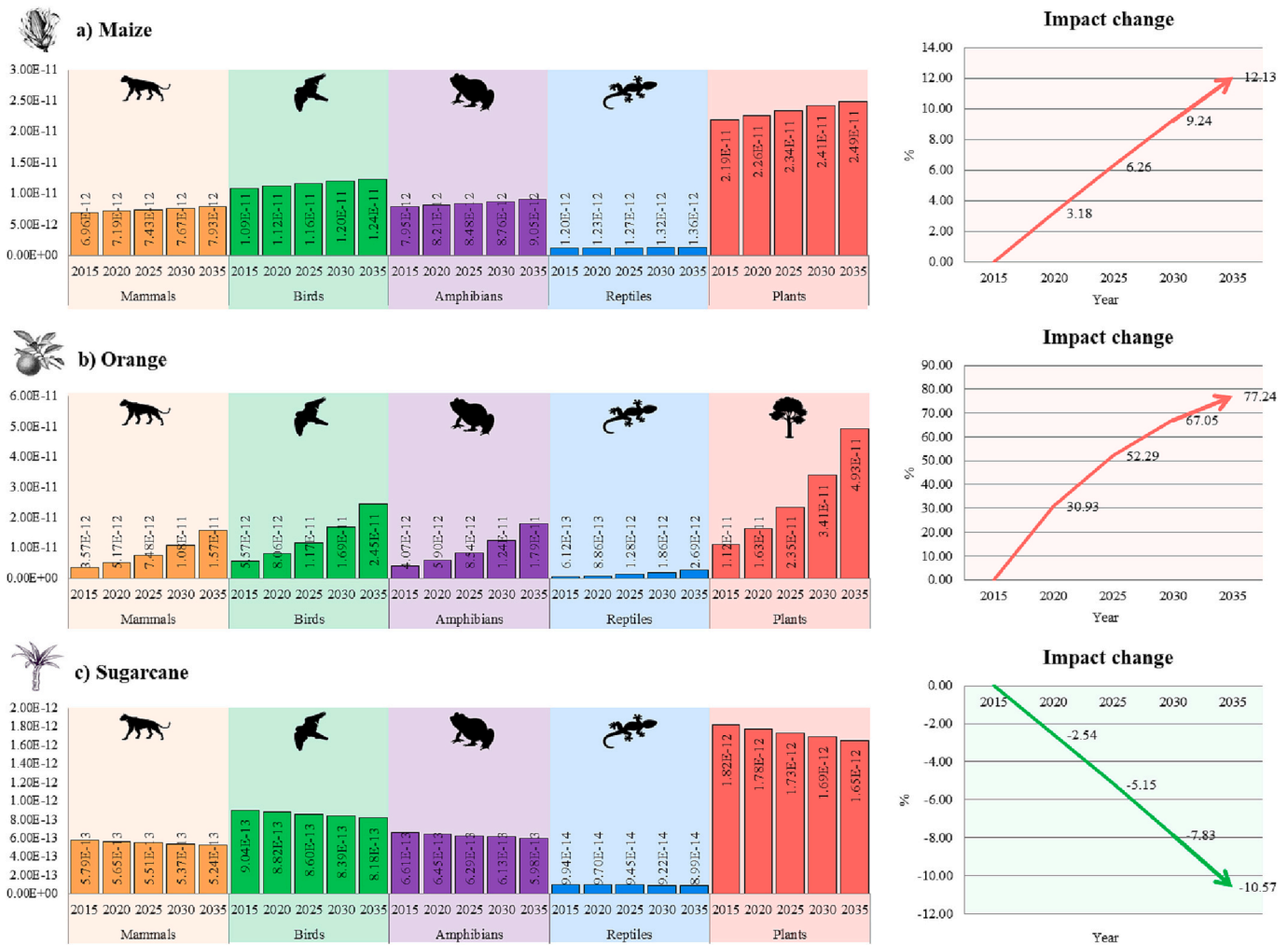


Fig. 8. Impact of potential biodiversity damage global (BD_{Global}), in potentially disappeared fraction (PDF) $\times$ year m^2 , for maize, orange and sugarcane crops in the Pantanal ecoregion for five taxa (mammals, birds, amphibians, reptiles, and plants). The information on the right side of the graphs indicate the increase or decrease (change) in the impacts of different scenarios.

impact was 30.93 %. In the prospective scenarios, the reduction may be 77.24 % between 2015 and 2035; between 2015 and 2020, the reduction in production was $5.08E+05$ kg of fruit (approximately 32 %) and $2.00E+04$ m^2 of area (approximately 2 %). Thus, the area/production ratio of orange changed from 0.81 in 2015 to 1.17 in 2020 and could reach 3.54 m^2 /production in 2035. The impacts on orange cultivation in the Pantanal were because of the significant reduction in production, even though there was reduction in the amount of cultivated area.

For sugarcane cultivation, increase in the impacts of damage to biodiversity between the five-year periods was also observed. In the estimated scenarios, the increase in impacts was 2.54 %, and for the prospective scenarios, the increase could be 10.57 % between 2015 and 2035. Between 2015 and 2020, there was an increase in production of $4.85E+07$ kg (approximately 1 %) and a reduction in area of $5.74E+06$ m^2 (approximately 1 %). Thus, the area/production ratio of sugarcane changed from 0.26 in 2015 to 0.276 in 2020 and could reach 0.331 m^2 /production of 1 kg of sugarcane in 2035. The increase in impacts on sugarcane cultivation in the Pampas has been because of the reduction in production rather than the reduction in the amount of cultivated area. Sugarcane cultivation was the only one to present increase in production and reduction in the damage to biodiversity in the Pantanal ecoregion.

4. Discussion

4.1. Agriculture, biodiversity damage, and preservation of ecoregions

LCA analysis allowed partial analysis of the impacts on biodiversity of important agricultural commodities in Brazil. Overall, for the crops in which yield gains were obtained, lesser BD were estimated. On the other hand, croplands replaced huge extensions of wild vegetation in Brazil. From 1985 to 2017, 11 % of the natural vegetation depleted for other land covers human usage, mostly in the Atlantic Forest, Cerrado, and Pampas (Sousa Jr et al., 2020).

In the same period, it was estimated that 63.5 % of the territory did not change its general composition—this does not imply that there were no degradation processes or no regeneration of the cover; changes in its structure and/or composition associated with the extraction of wood, products, or forest fires may have occurred, but were undetected at the national level. The largest amount of areas kept unchanged was in the Amazon ecoregion ($3.42E+15$ m^2), followed by the Cerrado ($1.00E+15$ m^2), the Atlantic Forest ($4.76E+11$ m^2), the Caatinga ($3.92E+11$ m^2), the Pantanal ($6.30E+10$ m^2), and the Pampas ($5.70E+10$ m^2). However, the remaining 36.5 % of the territory ($3.11E+15$ m^2) showed changes in land use, in which 33 % of the entire altered area was because of natural area loss ($1.02E+15$ m^2). Most of the area loss occurred in the Amazon (41.8 %) and the Cerrado (33.8 %), later in the Caatinga (10.7 %), in the Atlantic Forest (9.2 %), in the Pampas (2.9 %), and in the Pantanal (1.6

%). The change in land use in areas that were no longer natural was responsible for 31.6 % of the total change ($9.83\text{E}+11 \text{ m}^2$), mainly because of the replacement of pastures with agriculture, especially in the Atlantic Forest ecoregions (40, 9 %) and Cerrado (36.1 %) followed by Caatinga (13.7 %), Amazonas (6.6 %), Pampas (2.3 %), and Pantanal (0.3 %) (Sousa Jr et al., 2020).

It is known that the most devastated ecoregion in Brazil is the Atlantic Forest (Freitasa et al., 2010; Martins, 2011; MMA, 2022), as it is where most of its population and economic activities are found (INCT, 2022). It contains 72 % of the population and 70 % of the country's GDP (SOS Mata Atlântica, 2022). However, the ecoregions that have lost the most natural areas recently have been the Amazon (INPE, 2022; Silva Jr. et al., 2021; Trancoso, 2021) and the Cerrado (INPE, 2022; Strassburg et al., 2017; WWF, 2022). The two ecoregions are the main agricultural frontiers of the country (Rada, 2013; Rocha et al., 2011; Sauer, 2018; Stabile et al., 2020). Even the emergence of new cities because of the movement of people to these regions has been related to agriculture and livestock (Rada, 2013; Sauer, 2018; Carrero et al., 2020).

Not only in Brazil, but all over the world the agriculture has been the activity responsible for much of the natural area loss through land use change (Aznar-Sánchez et al., 2019), some important examples are: rice and oil palm production in southwest Asia (Gandhi and Jones, 2019; Pendrill et al., 2019; Richards and Friess, 2016); sorghum and oil palm production in Mexico and Central America (Bonilla-Moheno and Aide, 2020; Portillo-Quintero and Smith, 2018); the production of maize, cocoa, and forest extraction for charcoal in Africa (Carodenuto, 2019; Daggart et al., 2020); the need to expand food production because of global population growth (Bologna and Aquino, 2020; Mahapatra and Kant, 2005; Theurl et al., 2020) and the development and expansion of emerging economies and a global growing demand for energy (Afawubo and Noglo, 2019; Culas, 2012; FAO and UNEP, 2020) are among the main reasons we are seeing natural areas turn into monoculture fields (Antonarakis et al., 2022).

However, agricultural areas already installed can mitigate damage to biodiversity and contribute to the preservation of wild species if used in full consciousness of their functions and impacts. Improved management and use of agricultural areas can increase production (Finger et al., 2019; Rööß et al., 2018; Tilman et al., 2002) and the number of products grown in the same location (Kichel et al., 2014; Lemaire et al., 2014; Rodrigues et al., 2021), and thus minimize the pressure that agriculture has generated on still natural areas (Kichel et al., 2014; Phalan et al., 2011). Some examples of already consolidated activities that can increase the harmony between satisfactory agricultural production and biodiversity conservation include i) crop-livestock-forest integration (ILPF): a sustainable production strategy that includes farming, livestock, and forestry activities carried out in the same area in intercropping, aiming at synergistic effects between the components of the agroecosystem (Balbino et al., 2011). The strategy comprises four integrated agricultural systems: (a) crop-livestock; (b) crop-forest; (c) forest-livestock; and (d) crop-livestock-forest (Valani et al., 2020). In 2018, adherence to this practice reached 15 million hectares in Brazil, an eight-fold increase in thirteen years (Telles et al., 2021); ii) Integrated pest management (IPM): approach that aims to reduce pesticide use, by combining plant protection strategies and taking into account economic damage that shows that plants can tolerate injuries without compromising production (Chendler et al., 2011). The IPM concepts include the appropriate selection of control measures that predict economic benefits, taking into account decision rules on selection of control action, environmental benefits and damage to species that are not pests, but integrate agroecosystems (Dara, 2019; Kogan, 1998; Peterson et al., 2018); the use of ducks (Aigamo) in organic rice fields: this is a rice cultivation system that relies on ducks to eat insects and weeds. After planting the seedlings, the ducks are released into the rice fields and kept in place during the growing season (Asano et al., 1999). The presence of ducks in the rice fields eliminates the need to use inputs (pesticides and fertilizers), in addition to oxygenating the water, a clear example of the

interaction of ecosystem services with agriculture (Li et al., 2019; Per-nollet et al., 2015; Xu et al., 2017).

However, these and other techniques for the preservation of biodiversity can be efficient only if supported by legislative actions that guarantee the legal preservation of natural areas from illegal actions such as deforestation, extraction of organic products, or exploitation of mineral resources, while maintaining agricultural production and economic development (Lucas et al., 2021).

For this purpose, Brazil has its own Forest Code—Law No. 12.651/2012 (Brazil, 2012)—which aims to guarantee the presence of natural vegetation in all rural properties and strategic locations for the preservation of biodiversity and natural resources in general. The Art. 1-A, Law No. 12,651/2012, establishes the rules for the protection of vegetation, for permanent preservation areas, and for legal reserve areas. To this end, § 1º (nº. 1) affirms Brazil's sovereign commitment to preserving its forests and other forms of native vegetation, as well as biodiversity, soil, water resources, and the integrity of the climate system. In contrast, § 2º (nº. 2) deals with the importance of the strategic role of agricultural activity and the role of forests and other forms of native vegetation in ensuring sustainability, economic growth, improving the quality of life of the Brazilian population, and the country's presence in national and international food and bioenergy. The Art. 12 states that every rural property must maintain an area with native vegetation cover, as a legal reserve, the minimum percentage of preserved area is defined according to the location of the property. I—located in the *Amazônia Legal* (Legal Amazon): i) 80 %, in the property located in a forested area; 35 %, in the property located in the Cerrado area; c) 20 %, in the property located in the area of general fields. II—located in other regions of the country: 20 %.

Although the country has strict environmental legislation, we have seen deforestation and forest fires in Brazil remain unresolved (Oliveira et al., 2022; Silva et al., 2021), possibly because of inefficient enforcement in the implementation of their own measures (Coelho-Junior et al., 2022; Reydon et al., 2020). Only in the period from 2013 to 2019, the deforested area was $2.00\text{E}+13 \text{ m}^2$, it is estimated that about 90 % has been directly promoted by agriculture (Dummett et al., 2021). Most of the converted natural areas were in the Amazon and Cerrado ecoregions (Dummett et al., 2021; INPE, 2022). However, according to the study by Rajão et al. (2020), who evaluated illegal deforestation in Brazil from 2008 to 2020, only 2 % of properties cause 62 % of potentially illegal deforestation, which is concentrated in the Cerrado and Amazon ecoregions. Although a minority of farmers did not respect the legislative measures, this information shows that measures that ensure compliance with the laws must be a real priority in the preservation of natural areas that legally must be protected.

The loss of natural vegetation is not restricted to Latin America, including Brazil; the global deforestation of tropical forests from 2013 to 2019 was approximately $7.68\text{E}+11 \text{ m}^2$, with 60 % ($4.61\text{E}+11 \text{ m}^2$) resulting from the advance of agriculture, 69 % of it carried out illegally. This is notably true in countries such as Mexico, Argentina, Madagascar, the Democratic Republic of Congo, and Indonesia, in addition to Brazil, which together account for more than half of the global loss of tropical forests (Dummett et al., 2021).

A permanent increase in the loss of global natural areas has been observed in just a few years, and if this trend continues, the next few decades could present huge prospects of deforestation and species loss never seen before. Therefore, researchers and farmers urgently need to look for and implement measures that improve the harmony between agriculture and the preservation of natural areas and biodiversity. The technological development of agricultural activities that ensure increase in production while maintaining quality must be the key strategy to reduce the competition between agriculture and the preservation of biodiversity.

4.2. Technological development in agriculture as a means of preserving species

Technical agricultural development through better use of agricultural lands already occupied is essential for economic and social development, preventing the destruction of natural areas (Lucas et al., 2021). In this way, the results from this study allow us to argue about some of the aspects that, in the future, may help preserve wild species.

First, most of the scenarios evaluated suggested reductions in the impacts of damage to biodiversity by considering the crops individually. Paradoxically, many of these scenarios are from the ecoregion highly deforested, the Atlantic Forest (SOS Mata Atlântica, 2022) or are currently the most deforested: the Amazon and Cerrado (INPE, 2022). In general, the agriculture practised in these ecoregions involves huge amounts of investments received in recent years, mostly because of higher international commodity prices also catapulted by the high relation between the US dollar and the Brazilian real. These investments led to soaring levels of mechanization (Instituto Brasileiro de Geografia e Estatística-IBGE, 2017) and inputs—pesticides (Instituto Brasileiro de Geografia e Estatística-IBGE, 2022a) and fertilizers (Instituto Brasileiro de Geografia e Estatística-IBGE, 2022b).

Innovative technologies may also mitigate damages, as the ILPF practice, which encompassed $1.15 \times 10^{11} \text{ m}^2$ in 2016—roughly 87 % in the south, southeast, and central-west regions—regions that include extensions of the Amazon, Atlantic Forest, and Cerrado ecoregions (Embrapa, 2017). Concentrating the production of grains, cattle, and timber in the same field may also enhance system and productivity, along with better resource usage.

The data presented here evidences the importance of pursuing improvements in cultivation practices to guarantee the preservation of biodiversity in these megadiverse environments. In addition, the high demands of the world's growing population will have to be met through increasingly smaller land areas per capita (Theurl et al., 2020). The Food and Agriculture Organization—FAO (FAO, 2015)—propose that increasing agricultural productivity is a key element for reducing environmental stress, poverty and satisfying the high global demand for agricultural products. Accessing agricultural technology and information and targeting them is the way to correlate increased productivity with reduced impacts (Cui et al., 2018).

Second, the reduction of agricultural areas just contributes to mitigating impacts when production volumes are simultaneously maintained or increased at the point of reducing the area/production ratio (m^2/yield). Otherwise, the impact of reductions is illusory, because the areas continue to be deforested and affect the sustainability of production systems, compromising economic and social aspects (Andrée et al., 2019; Kamble et al., 2020; Sarkaret et al., 2021). As a consequence, other regions, in Brazil or in other countries, may be deforested to increase production and increase participation in the international market of such commodities. Both these increases favor the loss of natural areas (Andrée et al., 2019; Stern et al., 1996).

Therefore, the results showed indicate that the preservation of natural areas and native species depends directly on the possible increase in yields, which could be achieved by properly using the natural resources guided by scientific knowledge (Lamb et al., 2016). If this becomes true, together with the elaboration and fulfillment of legislative measures, changes in consumption habitats, etc., may also increase the likelihood of changing the harmful path of biodiversity loss worldwide.

Third, the taxa most affected were plants and birds, as observed by Green et al. (2005), Khoury et al. (2022), Lucas et al. (2021); Stanton et al. (2018). As smaller species, the existence and conservation of parks and reserves may work as palliative measures to partially preserve them (Pringle, 2017; Talal and Santelmann, 2019; Xie and Bulkeley, 2020). The US Conservation Reserve Program (CRP) offers farmers monetary incentives for the conversion of highly erodible farmland into wildlife habitat. This program led to positive effects on the conservation of a variety of birds across the North American continent (Blank, 2013;

Haegen et al., 2015).

Other small size animal groups affected by the elimination of native vegetation, such as amphibians and reptiles, may also benefit from the increase in conservation areas (Almeida-Gomes et al., 2014; Böhm et al., 2013; Cox et al., 2022; Halstead et al., 2022). A successful palliative measure for amphibian conservation was relieving captive frogs from Serra Nevada species raised in San Francisco Zoo (Halstead et al., 2022). The total extinction of two lizard species from the Indian Ocean Islands was also prevented by the Park Australia breeding program (Andrew et al., 2018; Emery et al., 2021).

Regarding the preservation of mammals, including diverse-sized animals, a greater variety of area sizes with extensive amounts is necessary. In this case, national parks and broad species conservation programs are key actions (Baisero et al., 2020; Pacifici et al., 2020; Smith and Peterson, 2021).

Fourth, the creation of impact assessment guides, such as the Global Guidance for Life Cycle Impact Assessment indicator (UNEP, 2016) is important to compare impacts on different ecoregions of the planet.

Finally, this study reinforces that agricultural technological development may mitigate biodiversity loss. It is also important to highlight that this development can be achieved by sustainable practices such as soil conservation measures, cover crops, integrated pest management, precision agriculture, and biological input usage in opposition to simply increasing chemical inputs such as fertilizers, machinery, and pesticides, which can compromise other LCA indicators. Efforts to fulfill and continuously improve the Brazilian Low-Carbon Agriculture Program can be a proper starting point (Carauta et al., 2021).

To date, bio inputs are in extremely fast development in Brazilian agriculture, both in terms of inclusion of biological agents in traditional companies portfolios and the appearance of startups and “on farm” production. Legislative measures are essential to guarantee the preservation of biodiversity both when affected by agriculture and illegal deforestation. However, in recent years, not just the existence of laws, but also the accomplishment of oversight and punishment of illegal deforestation could be a global effort to guarantee the extremely endangered portion of biomes worldwide (Coelho-Junior et al., 2022).

Although variations on individual crops yield and cultivated areas may suggest improvement or degradation of biodiversity, other factors may be taken into account: replacement of the considered crop by other agricultural or cattle activities; periodic increasing or decreasing of investments boosted by commodity prices; recent extreme climatic events may also shrink yields, as was the case with drought impacts on the southern and northeastern regions, where Pampas and Caatinga are located. Hence, these analyses are dynamic and may be understood in this context.

4.3. Quality of indicators for prospective LCA

The indicators used in this study (Chaudhary et al., 2015; Chaudhary and Books, 2018), in addition to being recommended by UNEP/SETAC (UNEP, 2016), are seen as the most promising for LCA nowadays. However, for a prospective LCA study, some limitations become clear during the study period and some strict assumptions are deemed necessary. As they are were indicators for determining land use occupation, land use changes are not considered in the equations; so the impacts caused by deforestation are disregarded. When applying indicators in the prospective approach, depending on the purpose of the evaluation, the results may be less enlightening. When evaluating the functional unit by using $\text{m}^2/\text{production}$, if an increase in production and stability or a reduction in the cultivated area are in course, the crop improvement may be estimated. The prospective studies can then estimate the reduction of impacts through technical development. However, if the reduction of impacts is given only by the reduction of the cultivated area, prospective scenarios will just estimate the crop extinction in that ecoregion, which is unlikely to happen. Thus, the decreasing impact projection may not be credible.

The main goal of this paper was to show how important it was to have a set of indicators to measure and compare BD, highlighting the possibility of cross country studies. Furthermore, the results presented summarize an important issue for the construction of an economic model that incorporates environmental costs into agricultural production.

5. Conclusion

Some crucial issues concerning the impacts of BD on agricultural commodities were presented and discussed in this paper. We brought out an assessment of the way monocultures unconditionally endanger biodiversity. The contributions of this paper were measuring the impacts of the field crops under the Brazilian biomes and propose a methodological improvement for LCA methodology. The results have indicated distinct impacts on biodiversity caused individually by the crops in each biome.

Although the Cerrado is one of the ecoregions in which deforestation has advanced the most, it has shown the area/production ratio that has developed the most. In opposition, the Pampas and Caatinga, which are not seen as agricultural frontiers, had negative results, increasing their impacts. Although the most optimistic scenarios for the future have been those in regions considered agricultural frontiers, at the same time they are the regions where agriculture is more technologically developed: coffee production in the Atlantic Forest; coffee production in the Cerrado; and orange production in the Amazon. Although the indicators do not consider land use change, these results indicate that the technological development of agriculture can really contribute to the mitigation of BD impacts in the future.

Along with agricultural technological development, the implementation and inspection of legislative measures are essential to prevent natural areas and biodiversity from continuing to decline. Otherwise, in the future, we will see records of deforestation and the disappearance of species. Thus, we urgently need researchers, farmers, and policymakers to act to link development with conservation.

For this reason, we believe that the availability of the indicators used here is a great advance for the LCA methodology; however, they can and should still be improved. This also means including parameters in the CFs that allow considering and differentiating agricultural management techniques. We realize that these are among the next steps for its development.

CRedit authorship contribution statement

All the authors participated in all stages of research, from data preparation, conclusion and study writing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Information about determining the values and functional units of the estimated scenarios is available through supporting information.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2023.165762>.

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